Georgia Kriebel

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Honours CS and Physics student at the University of Toronto. I build full-stack systems, work on physics-based simulation at the Dynamic Graphics Project, and am drawn to problems where mathematical structure meets real constraints.

Projects

Ranking Application | *Python, React, Supabase* June 2026 – Present

- Built and deployed a ranking platform as sole developer, owning the React frontend, Python backend, and Supabase schema
- Grew the platform to 25+ active users, with an App Store release currently in preparation

Hardware Supply Chain Optimizer | *Python, React, Graph Algorithms* March 2026 – June 2026

- Developed full-stack features at an early-stage startup using graph algorithms to optimize supply chain decisions
- Engineered a React frontend over a Python optimization backend to surface routing and sourcing results

Unix Shell in C | *C, Linux Kernel* March 2026

- Implemented a custom Unix shell supporting basic file operations, multi-process execution pipelines, and diagnostics
- Managed low-level system processes and memory handling in C under Linux

Rescue Pet Adoption Simulator | *Java, Java Swing, Clean Architecture, Petfinder API* Sep 2025 – Dec 2025

- Built a Java Swing application promoting animal adoption via an immersive simulation and live data from the Petfinder API
- Applied Clean Architecture principles to maintain a decoupled, testable, and scalable codebase on a collaborative project

Automated Geospatial Itinerary Planner | *Python, Node.js, Flask, Gemini API, OpenRoute Service API* Oct 2025

- Leveraged the Google Gemini API to assist travelers in generating a personalized itinerary
- Integrated the OpenRoute Service API via Node.js and Python to process geospatial requests and calculate optimal travel paths

Algorithmic Team Optimization Engine | *Python, Pygame, Graph Theory* Feb 2025 – April 2025

- Developed Python algorithms utilizing Graph Theory to calculate statistically ideal team compositions
- Modeled attributes as a directed graph to optimize team compatibility, and created an interactive UI using the Pygame library

Education

University of Toronto, Toronto ON, Canada August 2024 – Expected Graduation June 2028

Honors Computer Science and Physics Undergraduate (Focus in Scientific Computing) | Recipient of Warren Goldring Residence Scholarship

- Web Developer, University of Toronto Robotics Association (UTRA)
- Founder and President, University of Toronto Crochet Collective
- Co-Founder and Vice President, University of Toronto Matcha Club
- Member of UTQC (University of Toronto Quantum Computing); Volunteer for CSSU (Computer Science Student Union)

Experience

Undergraduate Researcher, Dynamic Graphics Project (DGP) | *Prof. Eitan Grinspun's Group* September 2026 – Present

- Developing a physics-based simulation of cloth pleating and pinning on 3D body models for garment design
- Contributing to work the group is preparing toward a SIGGRAPH submission

Barista and Customer Service, Mic's Market (Pittsburgh, PA USA) | *Communication, Delegation* Summers 2025 & 2026

- Delivered efficient, friendly customer service under high-speed conditions, taking care of orders with accuracy

Study Abroad: CIEE Japan Program, Kyoto, Japan | *Leadership, Adaptability, Japanese* April 2023 – August 2023

- Completed a Japanese immersion program with a host family; assisted non-English speakers in Model UN conference settings

Skills

- **Languages:** Python, Java, C, Assembly, JavaScript, TypeScript, R, SQL (PostgreSQL), HTML
- **Frameworks & Tools:** React, Node.js, Flask, Supabase, Java Swing, Pygame, Git, Linux
- **Hardware:** Arduino, VEX

Highlighted Courses

University of Toronto Current

CSC263 (Data Structures & Analysis), CSC209 (Software Tools & Systems Programming), CSC311 (Intro to Machine Learning), CSC367 (Parallel Programming), CSC317 (Computer Graphics), CSC364 (Foundations of Computer Security)